A practical path toward AI self-awareness

Research synthesis and experimental design for a 64 GB machine

Evidence reviewed through 11 October 2026. Recommendations are engineering hypotheses, not a validated recipe for sentience.

The recommendation

Build a persistent agent with a compact language model, a recurrent world/self model, a selective shared workspace, and a calibrated monitor that changes the agent's actions. Begin in a small simulated environment. Train the small modules on observed outcomes; use the language model for explanation and planning. Evaluate each proposed mechanism with controlled interventions.

There is no scientifically established neural structure that guarantees subjective experience, and no accepted test that certifies it in AI. The theory-derived indicator programme provides a research framework, conditional on contested assumptions about computation and consciousness. A recent revision appeared online in 2025 and in the June 2026 issue of Trends in Cognitive Sciences. [1,2]

Target	What you can establish
Functional self-awareness	The system predicts its capabilities, uncertainty, memory gaps and consequences of its own actions.
Access consciousness	Information is available for flexible reasoning, reporting and control, in the functional sense.
Phenomenal consciousness / sentience	There is something it feels like to be the system. Behavioral success or architectural similarity alone does not establish this.

What the 64 GB constraint means

The budget is enough for a useful inference prototype using pretrained components. It is not a realistic budget for recreating frontier-model pretraining. System RAM, GPU VRAM, and Apple-style unified memory have different performance implications. This brief assumes 64 GB of system or unified memory, and gives design reservations rather than measured benchmarks.

The first scientific question

Does a learned, causally active self-model improve calibration and adaptive control beyond ordinary memory, extra context, and extra compute? This is answerable. Whether that improvement entails experience remains an additional theoretical question.

What the research supports

Evidence	Result and limit
Consciousness indicators [1,2]	Architecture-based assessment can structure inquiry. Indicators are provisional and do not form a calibrated consciousness score.
Human theory testing [3]	The 2025 COGITATE comparison supported some predictions and challenged others for GNWT and IIT. It did not identify a settled blueprint.
LLM introspection [4]	Activation-injection experiments found limited access to internal states. Opus 4.1 detected injections around 20% of the time under the best reported protocol; failures and false interpretations were common.
LLM workspace [5]	July 2026 experiments identified verbalizable internal representations with workspace-like functions. The authors distinguish functional access from subjective experience.
Recurrent reasoning [6,7]	Huginn repeats a learned block in latent space. September 2026 LRT uses a small recurrent reasoner with a frozen decoder. Both concern reasoning, not consciousness.
Attention schema [8]	A small neural agent learned attention control using a description of attention. This supplies a useful intervention experiment rather than a universally accepted sentience test.
Biological naturalism [9]	Anil Seth argues consciousness may depend on properties of living systems. Functional similarity on digital hardware may therefore be insufficient.

These sources differ in evidential status. The COGITATE and attention-schema studies are peer-reviewed experiments; the indicators paper is a methodological synthesis; Huginn and LRT are machine-learning research papers, with LRT especially recent; Anthropic's workspace research is a developer-authored mechanistic study. Biological naturalism is an argued alternative, not proof that digital consciousness is impossible.

How to interpret a positive result

A successful ablation experiment can demonstrate a mechanism's causal contribution to a capability. Its relevance to consciousness still depends on whether the underlying theory is correct and whether that mechanism has been instantiated in the relevant way. Adding a module named "global workspace" is not itself evidence.

Choosing the components

Component	Recommended use	Boundary
Qwen3.5-9B [10]	Practical initial language/planning backbone; use 4-6 bit weights after quality checks.	Its hybrid recurrent/attention architecture does not imply sentience.
Qwen3.8-27B [11]	Larger current comparison model when latency permits; quantize to about 4 bits.	More parameters can leave less compute for experiments.
Huginn-3.5B [6]	Separate comparison of trained depth recurrence against token-based reasoning.	Depth recurrence is not persistent state across environmental episodes.
V-JEPA 2 / 2.1 [12,13]	Optional pretrained visual encoder; add an action-conditioned predictor for your environment.	Visual latent prediction does not automatically supply goals, a self-model or language alignment.
DreamerV3 [14]	Reference implementation for learning dynamics and evaluating plans through imagined outcomes.	Its reward-driven world model is not evidence of subjective feelings.
Jev [15]	Reference for fast typed decisions and uncertainty-aware routing.	Official material exposes a hosted API; I found no official downloadable weights or full architectural specification.

Jev and JEPA are different. JEPA is a predictive representation-learning approach. TypeSafe describes Jev as a model returning typed values and probabilities in parallel, trained with Reinforcement Learning for Calibrated Decisions. Those descriptions do not establish consciousness. Type safety prevents certain output-format failures; it does not make a factual judgment infallible. [12,15]

My choice under your constraint

Use Qwen3.5-9B first, and spend the saved resources on a small recurrent controller, grounded outcome data and repeatable tests. Keep Qwen3.8-27B as a quality comparison. If the main research question is internal recurrence, test Huginn separately; do not assume that repeating arbitrary layers in an ordinary pretrained transformer produces meaningful cognition.

V-JEPA 2.1 now includes an 80M-parameter ViT-B checkpoint and larger options in the official repository. For a first visual experiment, this gives a smaller entry point than the 300M ViT-L variant of V-JEPA 2. The small encoder is not the robot-trained action-conditioned model. [13]

A testable agent architecture

The following is my proposed design, inspired by the research rather than specified by it. A software implementation can establish functional organization; a later learned implementation can test stronger claims about neural representations.

Subsystem	State and responsibility
Perception	Encode observations; retain timestamps, visibility, noise estimates and provenance. Begin with structured observations; add video later.
World model	A recurrent latent belief state predicting observations and outcomes conditional on actions. Track uncertainty under partial observability.
Self-model	Predict task success, attention coverage, tool availability, memory retrieval reliability, resource costs and effects of interventions on the agent itself.
Shared workspace	Select a small set of currently relevant representations. Broadcast selected content to planner, memory, monitor and action policy; receive feedback that changes the next selection.
Memory	Store actual episodes separately from hypotheses, imagined futures and generated summaries. Retrieve by relevance; verify against original records.
Metacognitive monitor	Estimate reliability and disagreement. Cause abstention, inspection, additional reasoning, revised beliefs or different actions.
Planner / policy	Compare a small number of action sequences using predicted outcomes and uncertainty. Let the LLM propose and explain plans.
Scheduler	Run recurrent updates while active, trigger expensive inference selectively, and checkpoint enough state to resume after restart.

One cycle

Observe -> update world and self state -> select workspace contents -> predict alternatives -> estimate confidence -> act or inspect -> compare prediction with outcome -> record the grounded event -> update small learned components. Each next cycle depends on the previous state.

The world and self models should exchange information through an explicit recurrent state. Merely placing every module's prose in a shared prompt does not demonstrate a learned neural broadcast mechanism. For a stronger experiment, learn a fixed set of workspace slots with selection, read/write pathways and recurrent feedback, then ablate those pathways.

Starting dimensions, to be validated

Try 8-16 workspace slots of 256-512 dimensions and a 512-2,048 dimensional recurrent belief state. Start the learned predictor and monitor below 50M parameters combined. These are experimental starting points; no dimension, loop frequency or number of slots is a consciousness threshold.

Learning useful self-knowledge

Ground the self-model in predictions

A self-model should predict facts about the agent before it sees the outcome. Example: "With the left sensor occluded and my retrieved map 30 steps old, I have a 0.55 probability of reaching the target without inspecting again." Save that prediction, execute the episode and score it. Track outcomes by task, perturbation and model version.

Start with reversible changes: sensor dropout, delayed observations, unavailable tools, stale memory, changed action mappings, and small compute budgets. Include episodes where the environment changes but the agent does not, and vice versa. Avoid revealing the intervention label in the prompt when testing whether it is detected independently.

Train small modules first

For action-conditioned world prediction, encode the next observation with a frozen encoder and train a predictor from current latent state and action. Add multi-step evaluation on held-out trajectories. A JEPA-style objective needs a non-collapse strategy; use a frozen pretrained target initially rather than assuming a raw matching loss learns useful features.

Train success and reliability predictors using proper scoring losses, such as cross-entropy or Brier loss, on real episode outcomes. Train the attention monitor to predict whether selecting a source improves performance, then test its removal. Train memory reliability on actual retrieval correctness, including misleading near-matches.

Separate telemetry from introspection

Giving a model its RAM usage or task accuracy lets it reason about external telemetry. That is useful system self-monitoring, but does not show access to its own neural computation. Stronger introspection experiments require open weights, activation access, sham interventions and reports made before revealing changed outputs. Use a small PyTorch model for these experiments if the production runtime does not expose the needed activations.

Anthropic's May 2026 introspection-adapter work shows that self-reports of learned behaviors can be improved through training, including experiments with small Qwen3 models. The authors also report high false-positive rates. This motivates trainable self-reporting plus negative controls; it does not justify trusting every self-report. [16]

Persistence and internal activity

Implement continuous state maintenance while active, with slower planning and periodic replay/consolidation. An initial simulator might use 1-5 small state updates per second and one bounded LLM call every 5-30 seconds or on significant events. These are engineering settings, conditional on throughput. A human-like agent need not verbalize everything, and constant self-talk is not a validated requirement for consciousness.

Keep observations, inference and imagination separately marked. Never let repeated self-generated descriptions become evidence of lived experiences or successful actions.

A realistic 64 GB budget

All numbers below are decimal GB and are conservative design reservations for inference, not measurements on your hardware. They assume one language-model instance, one active session, an 8K-16K context, bounded video batches and a disk-backed replay store. Quantization metadata, unquantized layers and multimodal components change actual size.

Allocation	9B initial system	27B comparison
Language-model weights and optional vision weights	9 GB	18 GB
KV / recurrent inference caches	4 GB	4 GB
Execution buffers and temporary activations	6 GB	6 GB
Optional JEPA encoder and video working buffers	3 GB	3 GB
Small learned controllers / monitors	2 GB	2 GB
RAM-resident retrieval and replay cache	3 GB	3 GB
Operating system and other services	10 GB	10 GB
Contingency reservation	12 GB	12 GB
Total planned allocation	49 GB	58 GB

Unused capacity is valuable. A nominal 64 GB is about 59.6 GiB; distinguish the units when reading runtime logs. A 27B model can fit while still being too slow for the desired loop. Profile peak process memory and accelerator memory together, with the real context, sensory input and concurrency.

Weight and cache arithmetic

Packed weights alone need approximately N x b / 8 bytes for N parameters and b bits per parameter. Thus 9B at 6 bits is 6.75 GB before overhead; 27B at 4 bits is 13.5 GB. Model artifacts can be larger. A mixture-of-experts model normally needs all resident experts, not just the active parameter count.

For ordinary attention layers, KV bytes are approximately 2 x L x T x Hkv x D x bytes_per_element x batch. For Qwen3.5-9B, its 8 full-attention layers, 4 KV heads and 256-dimensional heads give 0.5 GiB of FP16 KV at 16,384 tokens and batch 1. This excludes its recurrent-state caches, buffers and visual token costs. [10,17]

Inference is not training

The table does not cover training. Mixed-precision Adam can require roughly 12-16 bytes per trainable parameter for weights, gradients and optimizer states, plus activations. Frozen-backbone inference, small-model training and QLoRA have different footprints. Do not assume that a 4-bit model which fits for inference fits for training. Reuse frozen components and train small predictors before attempting language-model adaptation.

For local serving, llama.cpp provides CPU, accelerator and mixed CPU/GPU execution. Use a separate instrumented PyTorch setup for neural interventions, and verify numerical differences after quantization. [18]

Tests that can change your conclusion

Evaluate capability and mechanisms separately from consciousness claims. Use held-out environments, matched compute budgets, repeated seeds and predeclared metrics. Confidence statements must be scored against outcomes.

Experiment	Measure and control
Remove the self-model	Measure success calibration, adaptation and inspection choices. Replace it with a same-size predictor lacking agent-specific features to control for capacity.
Cut workspace broadcast	Measure cross-module transfer and multi-step planning. Match information volume and compute in the replacement path.
Remove recurrence	Measure belief tracking through occlusions and delayed evidence. Compare against a feed-forward model with matched capacity/compute and an ordinary LLM-memory baseline.
Shuffle memory identities	Measure whether action planning respects actual episode provenance. Distinguish grounded records from plausible summaries.
Disable attention schema	Measure source selection under distraction and uncertainty. Keep sensor quality and raw input access comparable.
Intervene on internal activations	Test early identification with randomized shams, unrelated perturbations and false-positive accounting. Separate detection of processing from inference based on already-visible outputs.
Change agent / change world	Test whether the self-model attributes failure to its capability, information access or environment dynamics on held-out changes.

Metrics to log

Task success; negative log-likelihood and Brier score; reliability curves; false-positive introspection rate; unnecessary inspection rate; multi-step prediction error; adaptation delay; peak RAM/VRAM; wall-clock latency; total compute. Prefer environment-verified events over LLM-judged narratives.

A concrete first environment

Use a partially observed grid world with a limited field of view, a movable attention window, noisy cues, a remembered map and changing tool/action availability. Give the agent an objective such as delivering objects while choosing when to inspect. Ground truth lets you test world belief, attention, memory and self-capability independently. A simulated embodiment supports action-observation experiments; it does not establish that a biological body is unnecessary for experience.

A useful stopping rule

Advance a mechanism only if it improves a preregistered measure over a matched baseline and its targeted removal reverses the relevant benefit across held-out conditions. This is a research progression rule, not a consciousness threshold. No weighted sum of these results warrants an invented "percent sentient" score.

An eight-week research sequence

Stage	Deliverable	Decision gate
Weeks 1-2	Serve the 9B model; implement grounded episodes, structured self-state and a persistent scheduler. Benchmark memory and latency.	Establish a reproducible LLM + memory baseline.
Weeks 3-4	Train a small recurrent dynamics predictor and outcome-calibrated self-monitor in the simulator.	Improve held-out calibration or adaptation, beyond added compute.
Weeks 5-6	Introduce learned workspace selection/broadcast and an attention schema. Run controlled ablations.	Show pathway-specific benefit, not just a larger model benefit.
Weeks 7-8	Run controlled introspection probes on a smaller open model; compare recurrent reasoning alternatives; optionally add visual latents.	Replicate findings with shams, held-out interventions and repeated seeds.

This schedule is a suggested scope, not a promise of results. CPU-only training or unfamiliar tooling will extend it. The first milestone should be an auditable improvement in self-knowledge, not a charismatic personality.

The most promising narrow research direction

A small recurrent self-model, trained on hidden interventions to the agent's perception, memory and computation, that predicts its own failure modes and changes attention and planning through a shared workspace. Test whether those benefits remain after matching external information and compute. This is feasible to investigate under 64 GB and connects directly to self-awareness mechanisms without assuming that capability entails experience.

What would remain unresolved

Whether computational functionalism is true; whether neural recurrence in perception is required; whether workspace-like access entails experience; whether living self-maintenance matters; and whether a digital self-model has any felt perspective. These are substantive competing views, not small implementation gaps. [1,3,9]

What I would avoid as the primary objective

Training declarations of consciousness; rewarding distress or survival narratives; endlessly appending an inner monologue; using a Turing test as proof; assigning sentience from parameter count or RAM; and describing a predictive encoder as aware because it has a world model. If reports resembling suffering appear, treat them as uncertain evidence to investigate rather than a feature to optimize.

Hardware details that determine the implementation

Whether the 64 GB is ordinary RAM or unified memory; CPU model and memory bandwidth; GPU model and VRAM; operating system; and whether local training is required. These change the runtime, quantization, sensory throughput and feasible training strategy. They do not resolve the scientific question of experience.

Sources and reproducible starting points

Primary research, official model documentation and developer repositories. Accessed 11 October 2026. Papers and vendor claims are identified in the body; memory allocations and the proposed design are my estimates. Where full publisher text was inaccessible, only accessible metadata or abstracts were used.

[1] Butlin et al. (2023). Consciousness in Artificial Intelligence: Insights from the Science of Consciousness.
https://arxiv.org/abs/2308.08708

[2] Butlin et al. Identifying indicators of consciousness in AI systems. Online 2025; Trends in Cognitive Sciences 30(6), June 2026.
https://doi.org/10.1016/j.tics.2025.10.011

[3] COGITATE Consortium et al. (2025). Adversarial testing of global neuronal workspace and integrated information theories of consciousness. Nature.
https://doi.org/10.1038/s41586-025-08888-1

[4] Anthropic (29 Oct 2025). Signs of introspection in large language models / Emergent introspective awareness in LLMs.
https://www.anthropic.com/research/introspection

[5] Gurnee et al. (6 Jul 2026). Verbalizable Representations Form a Global Workspace in Language Models.
https://transformer-circuits.pub/2026/workspace/index.html

[6] Geiping et al. (2025). Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach. Huginn model card linked from paper.
https://arxiv.org/abs/2502.05171

[7] Chen & Fu (1 Sep 2026). Latent Recurrent Thoughts: Recurrent Refinement of Proposed Latents for Reasoning with Frozen LLMs. Recent preprint.
https://arxiv.org/abs/2609.01117

[8] Wilterson & Graziano (2021). The attention schema theory in a neural network agent. PNAS 118(33). Author-hosted paper.
https://grazianolab.princeton.edu/document/172

[9] Seth (2025). Conscious artificial intelligence and biological naturalism. Behavioral and Brain Sciences.
https://doi.org/10.1017/S0140525X25000032

[10] Qwen. Official Qwen3.5-9B model card.
https://huggingface.co/Qwen/Qwen3.5-9B

[11] Qwen. Official Qwen3.8-27B model card.
https://huggingface.co/Qwen/Qwen3.8-27B

[12] Assran et al. (2025). V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning.
https://arxiv.org/abs/2506.09985

[13] Mur-Labadia et al. (2026). V-JEPA 2.1. Checkpoint sizes in official repository.
https://github.com/facebookresearch/vjepa2

[14] Hafner et al. DreamerV3: Mastering diverse control tasks through world models. Nature 2025; arXiv version first posted 2023.
https://arxiv.org/abs/2301.04104

[15] TypeSafe AI. Introducing System One Models & Jev; API introduction. Official hosted interface.
https://docs.typesafe.ai/introduction

[16] Anthropic (May 2026). Introspection Adapters. Training approach, small-model experiments and false-positive limitations.
https://alignment.anthropic.com/2026/introspection-adapters/

[17] Qwen. Qwen3.5-9B configuration (attention-layer count, KV heads and dimensions).
https://huggingface.co/Qwen/Qwen3.5-9B/raw/main/config.json

[18] ggml-org. llama.cpp. Local inference and hardware backend documentation.
https://github.com/ggml-org/llama.cpp

[19] Anthropic. Jacobian lens reference implementation; examples use open-weight Qwen models. Fitting requires backward passes.
https://github.com/anthropics/jacobian-lens

[20] Mur-Labadia et al. V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning.
https://arxiv.org/abs/2603.14482